

FULL FARM MANAGEMENT SYSTEM (FFMS)

Objective

Design and develop an Intelligent Farm Management System (IFMS) using PHP as the backend language. The system should provide a centralized web-based platform for managing agricultural operations, including crop and livestock production, farm resources, irrigation, inventory, finances, labour, suppliers, sales, weather information, and AI-assisted decision-making.

1. Farm & Field Management

- Farm registration and profile management
- Multiple farms and field/plot management
- GPS/map-based farm locations
- Field size and boundary management
- Soil type and soil-condition records
- Crop rotation and field history
- Land-use planning

2. Crop Management

- Crop and variety registration
- Planting schedules
- Seed and seedling management
- Planting density and spacing
- Fertilizer and manure application records
- Pest and disease monitoring
- Crop growth monitoring
- Crop protection/spraying schedules
- Expected harvest dates
- Actual harvest records
- Yield per field/hectare

4. Irrigation & Water Management

- Irrigation system registration
- Drip, sprinkler and other irrigation systems
- Water-source management
- Irrigation scheduling
- Water consumption tracking
- Pump and equipment monitoring
- Irrigation-cost tracking
- Weather-based irrigation recommendations
- Alerts for irrigation problems

5. Livestock Management

- Animal registration
- Breed management
- Animal identification
- Birth and death records
- Vaccination schedules
- Disease and treatment records
- Feed management
- Weight/growth monitoring
- Breeding records
- Livestock production records
- Mortality analysis

6. Farm Inventory & Inputs

- Seeds and seedlings
- Fertilizers
- Chemicals and pesticides
- Animal feed
- Veterinary medicines
- Packaging materials
- Fuel
- Tools and equipment
- Stock levels and expiry dates
- Stock-in/stock-out records
- Low-stock alerts

7. Farm Equipment & Machinery

- Tractor and machinery registration
- Equipment allocation
- Maintenance schedules
- Fuel consumption
- Operating hours
- Repair history
- Equipment depreciation
- Maintenance reminders

8. Labour & Employee Management

- Farm-worker registration
- Employee roles
- Attendance

- Work allocation
- Daily task management
- Labour costs
- Productivity tracking
- Payroll integration
- Performance monitoring

9. Pest & Disease Management

- Pest and disease database
- Field scouting records
- Disease/pest reporting
- Recommended control measures
- Spraying schedules
- Chemical application records
- Treatment effectiveness tracking
- Disease outbreak alerts

10. Weather & Environmental Monitoring

- Current weather information
- Weather forecasts
- Rainfall monitoring
- Temperature monitoring
- Humidity
- Wind conditions
- Extreme-weather alerts
- Weather-based crop recommendations
- Historical weather analysis

11. Harvest Management

- Harvest scheduling
- Harvest quantities
- Field-by-field yield records
- Grade and quality
- Losses and wastage
- Post-harvest handling
- Storage records
- Packaging
- Actual versus expected yield analysis

12. Market & Sales Management

- Customer management
- Buyer registration
- Market-price monitoring
- Crop demand monitoring
- Sales orders
- Invoices
- Payments
- Market destinations
- Contract farming
- Supplier/buyer agreements
- Profitability by crop

13. Financial Management

- Farm income
- Farm expenditure
- Production costs
- Input costs
- Labour costs
- Fuel costs
- Equipment costs
- Loan tracking
- Budget management
- Profit and loss
- Return on investment
- Cost per hectare
- Cost per kilogram

14. Supplier & Procurement Management

- Supplier registration
- Supplier quotations
- Purchase orders
- Input procurement
- Supplier performance
- Purchase history
- Payment tracking
- Procurement approvals

15. Storage & Post-Harvest Management

- Warehouse management
- Cold-storage management
- Stock movement

- Produce grading
- Produce quality
- Storage duration
- Spoilage monitoring
- Inventory valuation
- Dispatch management
- Expected yield
- Expected harvest date
- Yield per hectare
- Production losses
- Factors affecting yield
- Expected revenue
- Historical farm data can be used to compare predicted yield against actual harvest results.
- Expected selling price
- Price trends
- Demand trends
- Best selling period
- Potential revenue
- Market risks

18. Farm Dashboard & Analytics

- Total farm acreage
- Active crops
- Livestock numbers
- Current crop stages
- Expected harvest
- Actual production
- Farm expenses
- Revenue
- Profit
- Inventory levels
- Weather conditions
- Pest/disease alerts
- Irrigation status
- Market prices

19. Notifications & Alerts

- Planting dates
- Fertilizer application
- Spraying schedules

- Irrigation schedules
- Vaccinations
- Disease outbreaks
- Low inventory
- Expiring chemicals
- Equipment maintenance
- Expected harvest
- Weather risks
- Market opportunities

20. Reports & Analytics

- Crop production reports
- Yield reports
- Livestock reports
- Financial reports
- Expense reports
- Sales reports
- Inventory reports
- Labour reports
- Irrigation reports
- Pest/disease reports
- Equipment reports
- Profitability reports
- Farm performance reports

21. Security & User Management

- Administrator accounts
- Farm-owner accounts
- Farm-manager accounts
- Agronomist accounts
- Worker accounts
- Accountant accounts
- Role-based access control
- Secure authentication
- Password encryption
- Audit logs
- Data backup
- Secure API communication

Suggested Technology Stack

Component	Technology
Frontend	React / HTML / CSS / JavaScript
Backend	PHP
Database	MySQL / PostgreSQL
Maps	Google Maps / OpenStreetMap APIs
Weather	Weather API
Notifications	SMS/Email APIs
Authentication	JWT / Session Authentication
Charts	Chart.js